Arrays and Hashing Fundamentals

LumoraAI Presentation Demo • 01 Data Structures Algorithms

Core concepts

• Arrays store elements in contiguous logical positions and provide constant-time indexed access when the index is known.

• Hash tables map keys to values through a hash function. Average-case lookup, insertion, and deletion are commonly modeled as O(1), while collisions require a collision-resolution strategy.

• A frequency map is useful when a problem asks how many times each value occurs. In Java this is commonly represented with HashMap.

Complexity reasoning

• A single pass through n elements is O(n). Sorting n elements with a comparison sort is typically O(n log n). Building a frequency map in one pass is O(n) average time with O(k) additional space, where k is the number of distinct keys.

Interview patterns

• Two-sum style problems can use a hash map to store previously seen values and check whether the complementary value exists. The main design trade-off is extra memory in exchange for avoiding a quadratic pair enumeration.

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